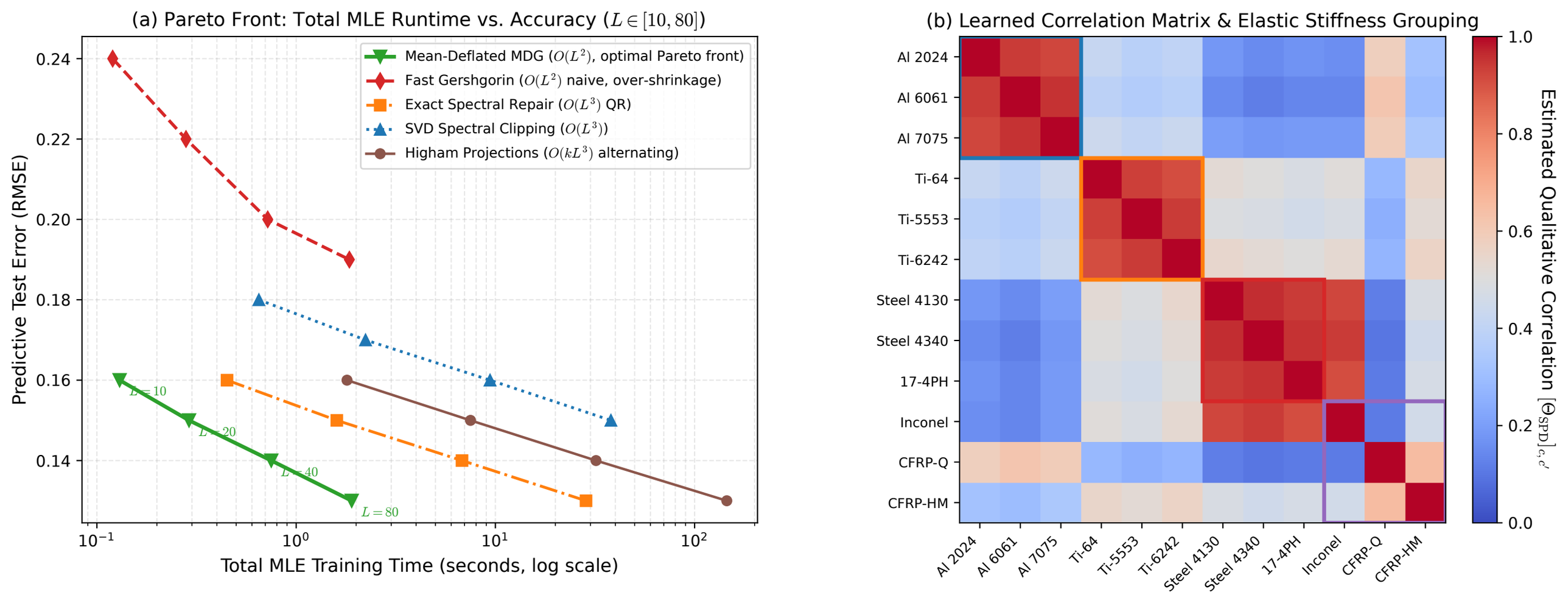
(a) Pareto Front: Total MLE Runtime vs. Accuracy ($L \in [10, 80]$)
Mean-Deflated MDG ($O(L^2)$, optimal Pareto front)
Fast Gershgorin ($O(L^2)$ naive, over-shrinkage)
Exact Spectral Repair ($O(L^3)$ QR)
SVD Spectral Clipping ($O(L^3)$)
Higham Projections ($O(kL^3)$ alternating)
$L = 10$
$L = 20$
$L = 40$
$L = 80$
Predictive Test Error (RMSE)
Total MLE Training Time (seconds, log scale)
(b) Learned Correlation Matrix & Elastic Stiffness Grouping
Al 2024
Al 6061
Al 7075
Ti-64
Ti-5553
Ti-6242
Steel 4130
Steel 4340
17-4PH
Inconel
CFRP-Q
CFRP-HM
Estimated Qualitative Correlation $[\Theta_{\mathrm{SPD}}]_{c,c'}$